

Basic Networking Skills for Digital Forensics

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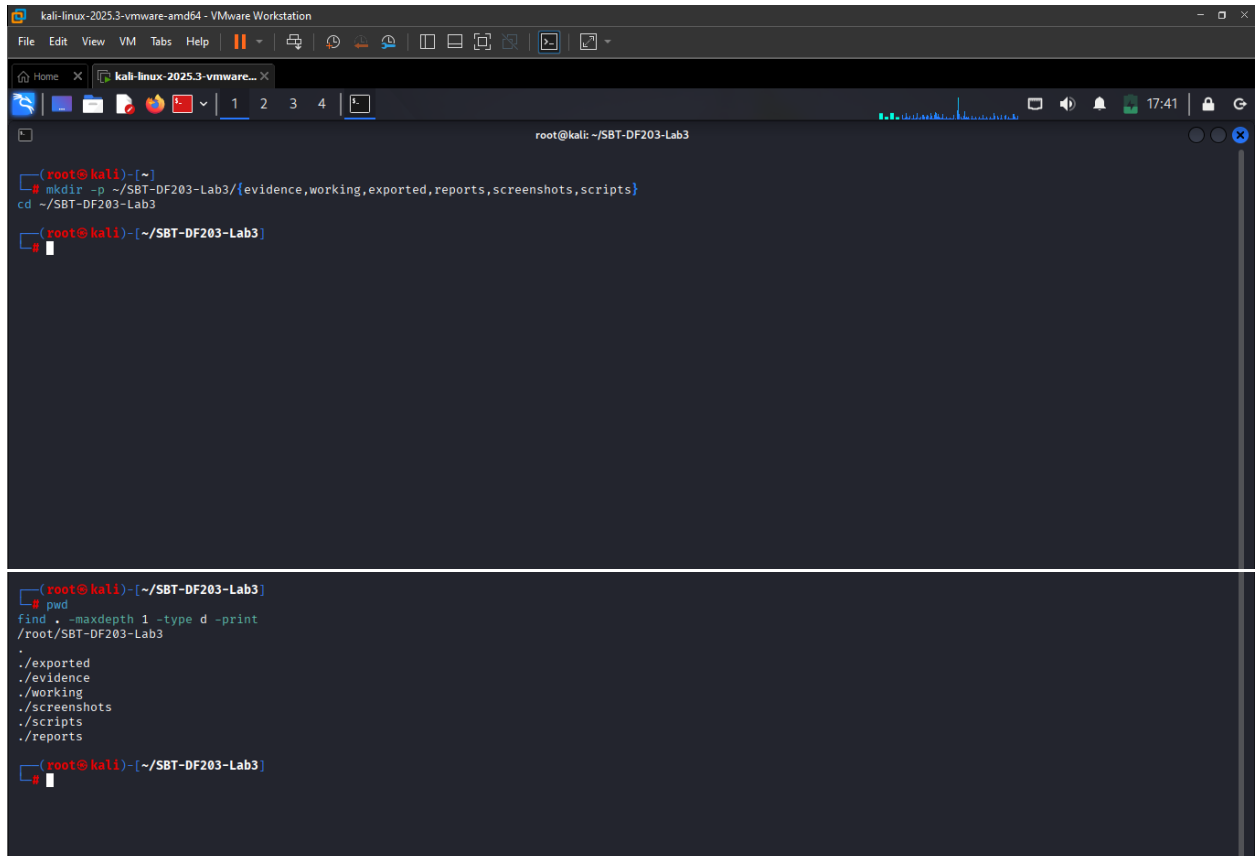
Fellowship in Web Application Security & Digital Forensics

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The objective of this lab was to analyse TCP connection behaviour, distinguish between normal and incomplete TCP handshakes, identify SYN packet indicators using Tshark, and perform a controlled loopback SYN simulation in a safe environment.

1. Folder structure Creation

A screenshot of a Kali Linux terminal window running inside a VMware Workstation. The terminal shows the user root at the kali machine, in the directory ~/SBT-DF203-Lab3. The user enters the command 'mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}' to create a directory structure. Then, they enter 'cd ~/SBT-DF203-Lab3' to move into that directory. Finally, they run 'find . -maxdepth 1 -type d -print' to list the created subdirectories, which outputs: './exported', './evidence', './working', './screenshots', './scripts', and './reports'.

```
(root@kali)~[~]
# mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab3

(root@kali)~/SBT-DF203-Lab3

(root@kali)~/SBT-DF203-Lab3
# pwd
# find . -maxdepth 1 -type d -print
/root/SBT-DF203-Lab3
.
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)~/SBT-DF203-Lab3
#
```

A folder structure was created with the above evidence using: `mkdir -p ~/SBT-DF203-Lab3/{evidence,working,exported,reports,screenshots,scripts}`
`cd ~/SBT-DF203-Lab3`

2. Apache Status up and running

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab3

(root@kali)~[~/SBT-DF203-Lab3]
# sudo apt update
# sudo apt install -y apache2 tshark wireshark python3-scapy
# sudo systemctl enable --now apache2
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.7 MB]
Get:4 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:5 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Fetched 76.4 MB in 48s (1,596 kB/s)
2131 packages can be upgraded. Run 'apt list --upgradable' to see them.
apache2 is already the newest version (2.4.68-1).
tshark is already the newest version (4.6.6-1).
tshark set to manually installed.
wireshark is already the newest version (4.6.6-1).
The following packages were automatically installed and are no longer required:
  gir1.2-girepository-2.0 libcrypto6 libjs-underscore libmbcrypto16 libraw23t64 openjdk-21-jre python3.13-venv
  imagemagick-7-common libcrypt-dev libjxr-tools libmpeg2utils-2.1-0t64 libtsk19t64 openjdk-21-jre-headless rpcsvc-proto
  libann0 libgavi-1 libjxr0t64 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs samba-ad-dc samba-ad-provision
  libarmadillo14 libgdal37 liblab-gamut1 libmplex2-2.1-0t64 libwiretap15 python3-gpg python3.13 samba-dsdb-modules
  libavc1394-0 libgirepository-1.0-1 liblqr-1-0 libpoppler147 libwmf-lite-0.2-7 python3.13
  libblosc2-4 libgpgmepp6t64 libmagickcore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev
  libcdt5 libjs-jquery-ui libmagickwand-7.q16-10 libpython3.13-dev libzxing3 python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Upgrading:
  libgpg-error-l10n python3-scapy

Summary:
Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 2129
Download size: 2,121 kB
Space needed: 786 kB / 55.8 GB available

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab3

imagemagick-7-common libcrypt-dev libjxr-tools libmpeg2utils-2.1-0t64 libtsk19t64 openjdk-21-jre-headless rpcsvc-proto
libann0 libgavi-1 libjxr0t64 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs samba-ad-dc samba-ad-provision
libarmadillo14 libgdal37 liblab-gamut1 libmplex2-2.1-0t64 libwiretap15 python3-gpg python3.13 samba-dsdb-modules
libavc1394-0 libgirepository-1.0-1 liblqr-1-0 libpoppler147 libwmf-lite-0.2-7 python3.13
libblosc2-4 libgpgmepp6t64 libmagickcore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev
libcdt5 libjs-jquery-ui libmagickwand-7.q16-10 libpython3.13-dev libzxing3 python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Upgrading:
  libgpg-error-l10n python3-scapy

Summary:
Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 2129
Download size: 2,121 kB
Space needed: 786 kB / 55.8 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 libgpg-error-l10n all 1.61-5 [119 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 python3-scapy all 2.7.0+dfsg1-1 [2,002 kB]
Fetched 2,121 kB in 5s (454 kB/s)
(Reading database... 385869 files and directories currently installed.)
Preparing to unpack .../libgpg-error-l10n_1.61-5_all.deb...
Unpacking libgpg-error-l10n (1.61-5) over (1.55-2)...
Preparing to unpack .../python3-scapy_2.7.0+dfsg1-1_all.deb...
Unpacking python3-scapy (2.7.0+dfsg1-1) over (2.6.1+dfsg1-1)...
Setting up python3-scapy (2.7.0+dfsg1-1)...
Setting up libgpg-error-l10n (1.61-5)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2025.3.2)...
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2

(root@kali)~[~/SBT-DF203-Lab3]

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
```

3. Downloading evidence

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3

(root@kali)~[~/SBT-DF203-Lab3]
# wget -O evidence/mySYNFloodCapture.pcap 'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap'
Prepended http:// to '/'
http://: Invalid host name.
--2026-09-11 17:48:55-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/lab_files/SYN_Flood/mySYNFloodCapture.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 260716 (255K) [application/octet-stream]
Saving to: 'evidence/mySYNFloodCapture.pcap'

evidence/mySYNFloodCapture.pcap 100%[=====] 254.61K 341KB/s in 0.7s

2026-09-11 17:48:57 (341 KB/s) - 'evidence/mySYNFloodCapture.pcap' saved [260716/260716]

FINISHED --2026-09-11 17:48:57--
Total wall clock time: 1.9s
Downloaded: 1 files, 255K in 0.7s (341 KB/s)

(root@kali)~[~/SBT-DF203-Lab3]
```

4. Sha256sum hash captured as:

14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5

```
(root@kali)~[~/SBT-DF203-Lab3]
# sha256sum evidence/mySYNFloodCapture.pcap | tee reports/syn_capture_sha256.txt
14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39baa55f5 evidence/mySYNFloodCapture.pcap
```

File details.

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3

(root@kali)~[~/SBT-DF203-Lab3]
# ls -lh evidence/mySYNFloodCapture.pcap
-rw-rw-r-- 1 root root 255K Sep 11 17:48 evidence/mySYNFloodCapture.pcap

(root@kali)~[~/SBT-DF203-Lab3]
#
```

5. Mini Evidence and Chain of Custody Worksheet

Field	My Entry
Case/lab identifier	SBT-DF203-Lab3-Idriss-Muhammad-Abdullahi
Trainee name	Idriss Muhammad Abdullahi
Date and time started	05/09/2026 00.00
Evidence file name(s)	normal_http.pcapng

Source or generation method	Apace SYN Flood
Original SHA-256	14765b029a72e9c41dd8b4d32f5b1d2c7d9efee0f084949151f183a39b aa55f5
Working-copy SHA-256	d4aabf495e8177d20cba1fbae08b2663176d449b526fbcf2cb48ebceb cd9b34a
Analysis workstation/V M	Kali Linux
Notes on any changes	No any modification that occurs

Part A. Establishing a Normal Handshake Baseline

Capturing normal HTTP traffic

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab3

(root@kali)~/~/SBT-DF203-Lab3
# sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w evidence/normal_http.pcapng 5
[1] 16631

(root@kali)~/~/SBT-DF203-Lab3
# Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'

[1] + done sudo tshark -i lo -f 'tcp port 80' -a duration:15 -w
(root@kali)~/~/SBT-DF203-Lab3
# curl --no-keepalive http://127.0.0.1/ >/dev/null
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 10703 100 10703 0 0 72381 0 Total Spent Left Speed
0

(root@kali)~/~/SBT-DF203-Lab3
#

```

Extracting TCP Handshake information

```

(root@kali)~/~/SBT-DF203-Lab3
# tshark -r evidence/normal_http.pcapng \
-Y 'tcp.flags.syn=1 || tcp.flags.fin=1' \
-T fields \
-e frame.number \
-e frame.time \
-e ip.src \
-e tcp.srcport \
-e ip.dst \
-e tcp.dstport \
-e tcp.flags \
| tee reports/normal_handshake_flags.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

(root@kali)~/~/SBT-DF203-Lab3
#

```

A normal HTTP connection was generated to the local Apache web server and captured using Tshark.

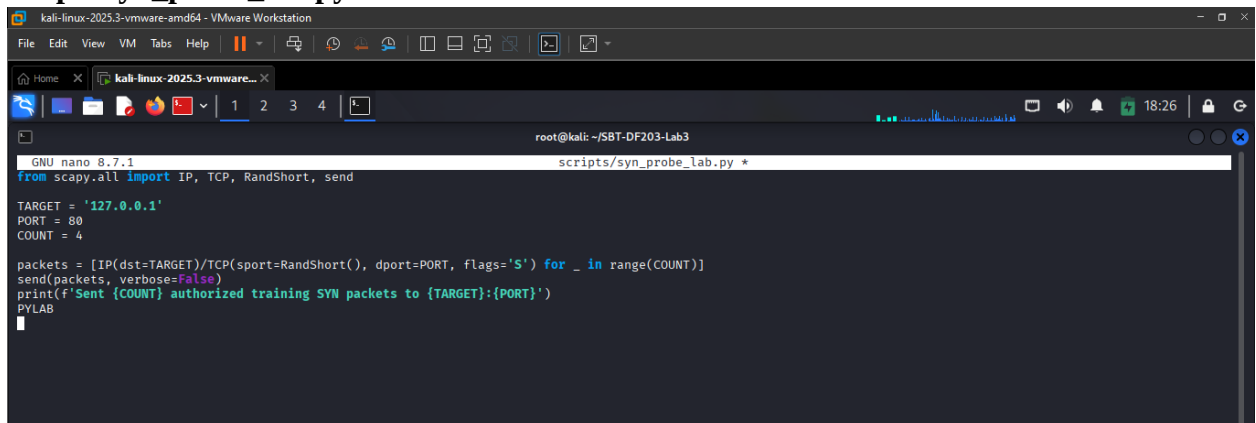
Findings

- A standard TCP three-way handshake was observed.:
 1. SYN from client
 2. SYN-ACK from server
 3. ACK from client
- HTTP communication followed the successful connection establishment.

The capture demonstrated expected TCP behaviour and provided a baseline for comparison against suspicious traffic.

Part B: Conducting the Bounded Loopback Simulation

Created a py load using: nano scripts/syn_probe_lab.py instead of cat scripts/syn_probe_lab.py

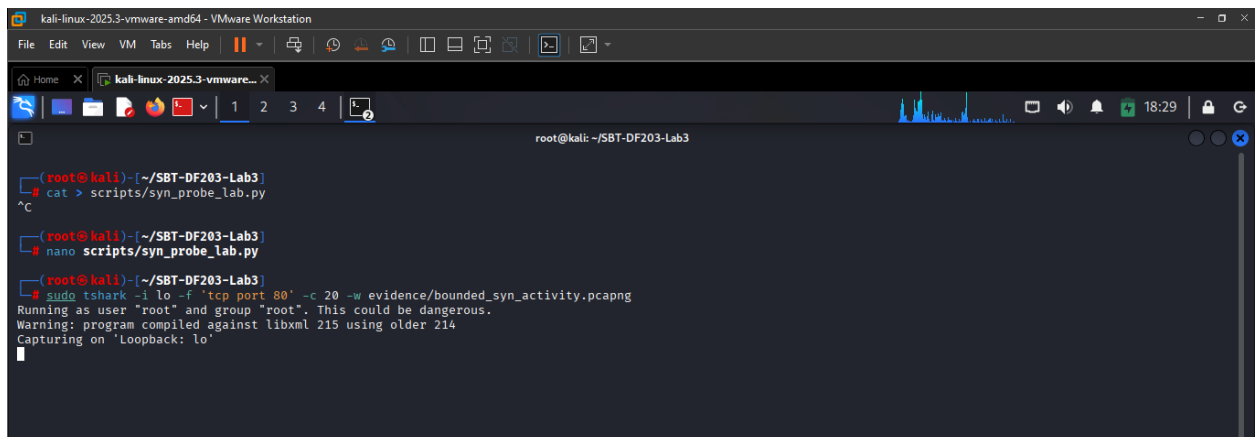


```
GNU nano 8.7.1 scripts/syn_probe_lab.py
from scapy.all import IP, TCP, RandShort, send

TARGET = '127.0.0.1'
PORT = 80
COUNT = 4

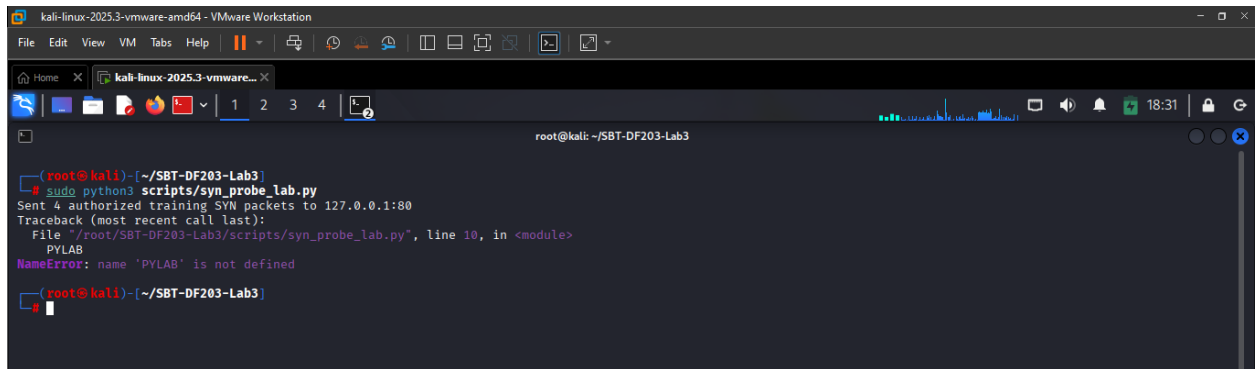
packets = [IP(dst=TARGET)/TCP(sport=RandShort(), dport=PORT, flags='S') for _ in range(COUNT)]
send(packets, verbose=False)
print(f'Sent {COUNT} authorized training SYN packets to {TARGET}:{PORT}')
PYLAB
```

Terminal 1



```
(root@kali) ~/SBT-DF203-Lab3
cat > scripts/syn_probe_lab.py
^C
nano scripts/syn_probe_lab.py
sudo tshark -i lo -f 'tcp port 80' -c 20 -w evidence/bounded_syn_activity.pcapng
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Capturing on 'Loopback: lo'
```

Terminal 2



```
(root@kali)-[~/SBT-DF203-Lab3]
# sudo python3 scripts/syn_probe_lab.py
Sent 4 authorized training SYN packets to 127.0.0.1:80
Traceback (most recent call last):
  File "/root/SBT-DF203-Lab3/scripts/syn_probe_lab.py", line 10, in <module>
    PYLAB
NameError: name 'PYLAB' is not defined
(root@kali)-[~/SBT-DF203-Lab3]
```

This is the successful result from terminal 2:

Sent 4 authorized training SYN packets to 127.0.0.1:80

A Scapy script was executed to transmit four authorized SYN packets to the local Apache service on 127.0.0.1:80.

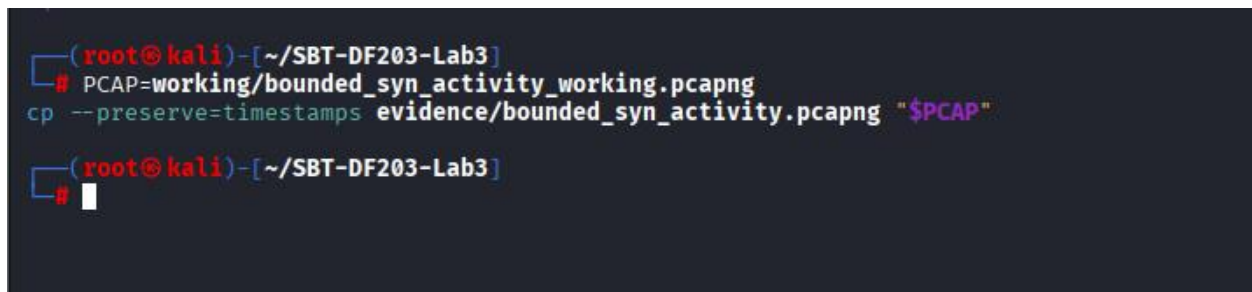
Findings

- Four SYN packets were successfully sent.
- The activity was isolated to the loopback interface.
- The script terminated automatically after transmitting the configured packet count.
- No external systems were targeted.

The simulation safely reproduced SYN connection attempts for forensic analysis without causing service disruption.

Part C: Identifying SYN Indicators with tshark

1. Analysis copy



```
(root@kali)-[~/SBT-DF203-Lab3]
# PCAP=working/bounded_syn_activity_working.pcapng
cp --preserve=timestamps evidence/bounded_syn_activity.pcapng "$PCAP"
(root@kali)-[~/SBT-DF203-Lab3]
#
```

Original and working Hashes matches

d4aabf495e8177d20cba1fbae08b2663176d449b526fbcf2cb48ebcebcd9b34a
(evidence/bounded_syn_activity.pcapng)

d4aabf495e8177d20cba1fbae08b2663176d449b526fbcf2cb48ebcebcd9b34a
(working/bounded_syn_activity_working.pcapng)

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware... X
1 2 3 4 5
root@kali: ~/SBT-DF203-Lab3
(root@kali)~[~/SBT-DF203-Lab3]
# sha256sum evidence/bounded_syn_activity.pcapng "$PCAP" \
| tee reports/bounded_capture_hashes.txt
d4aabf495e8177d20cba1fbae08b2663176d449b526fbcf2cb48ebcebcd9b34a evidence/bounded_syn_activity.pcapng
d4aabf495e8177d20cba1fbae08b2663176d449b526fbcf2cb48ebcebcd9b34a working/bounded_syn_activity_working.pcapng
(root@kali)~[~/SBT-DF203-Lab3]
#
```

2. Initial SYN Packets

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home X kali-linux-2025.3-vmware... X
1 2 3 4 5
root@kali: ~/SBT-DF203-Lab3
(root@kali)~[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.seq \
| tee reports/initial_syns.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1 1789165854.973129559 127.0.0.1 28026 127.0.0.1 80 0
4 1789165854.976008952 127.0.0.1 51113 127.0.0.1 80 0
7 1789165854.977707726 127.0.0.1 16426 127.0.0.1 80 0
10 1789165854.978640376 127.0.0.1 1280 127.0.0.1 80 0
(root@kali)~[~/SBT-DF203-Lab3]
#
```

SYN packet lists

3. SYN Attack Responses

```
(root@kali)~[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==1' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.ack \
| tee reports/syn_ack_responses.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2 1789165854.973770473 127.0.0.1 80 127.0.0.1 28026 1
5 1789165854.976629504 127.0.0.1 80 127.0.0.1 51113 1
8 1789165854.977757005 127.0.0.1 80 127.0.0.1 16426 1
11 1789165854.978688080 127.0.0.1 80 127.0.0.1 1280 1
(root@kali)~[~/SBT-DF203-Lab3]
#
```

4. Resets and final ACK candidates

```
(root@kali)-[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -Y 'tcp.flags.reset==1 || (tcp.flags.ack==1 && tcp.len==0)' -T fields \
-e frame.number -e frame.time_epoch -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e tcp.flags \
| tee reports/ack_reset_candidates.tsv

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
 2 1789165854.973770473 127.0.0.1 80 127.0.0.1 28026 0x0012
 3 1789165854.974402830 127.0.0.1 28026 127.0.0.1 80 0x0004
 5 1789165854.976629504 127.0.0.1 80 127.0.0.1 51113 0x0012
 6 1789165854.976685864 127.0.0.1 51113 127.0.0.1 80 0x0004
 8 1789165854.977757005 127.0.0.1 80 127.0.0.1 16426 0x0012
 9 1789165854.977819267 127.0.0.1 16426 127.0.0.1 80 0x0004
11 1789165854.978688080 127.0.0.1 80 127.0.0.1 1280 0x0012
12 1789165854.978754277 127.0.0.1 1280 127.0.0.1 80 0x0004

(root@kali)-[~/SBT-DF203-Lab3]
#
```

tshark filters were used to extract SYN packets, SYN-ACK responses, and ACK/RST behaviour.

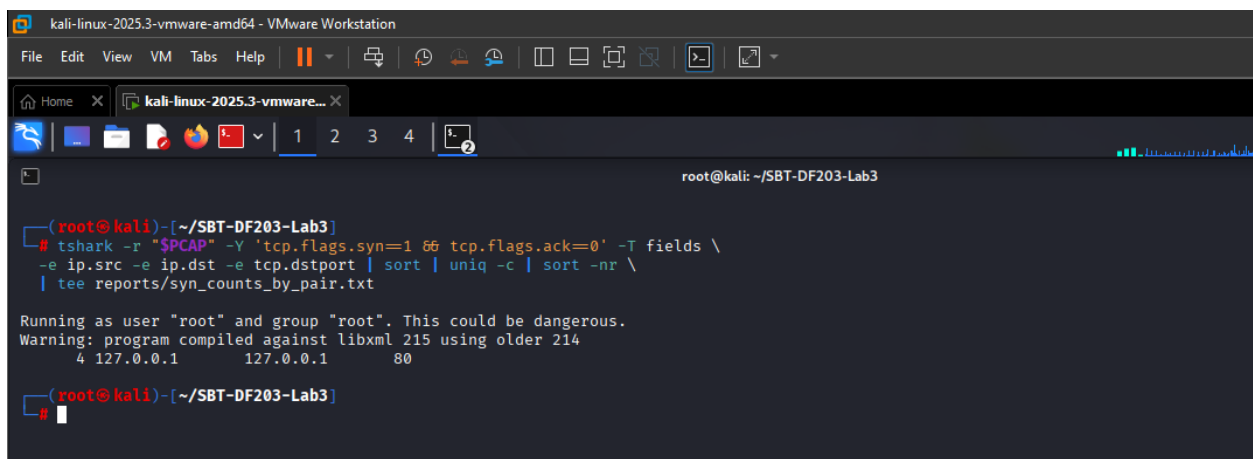
Findings:

- Initial SYN packets were identified and listed.
- Corresponding SYN-ACK responses were observed from the Apache server.
- ACK and/or RST responses were captured depending on connection completion behaviour.
- Packet timestamps supported accurate event reconstruction.

The extracted packet data clearly showed the relationship between connection initiation attempts and server responses.

Part D: Quantify and Correlate the Activity

1. Counting of the SYN Packets



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
Home X
1 2 3 4
root@kali: ~/SBT-DF203-Lab3

(root@kali)-[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields \
-e ip.src -e ip.dst -e tcp.dstport | sort | uniq -c | sort -nr \
| tee reports/syn_counts_by_pair.txt

Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
 4 127.0.0.1 127.0.0.1 80

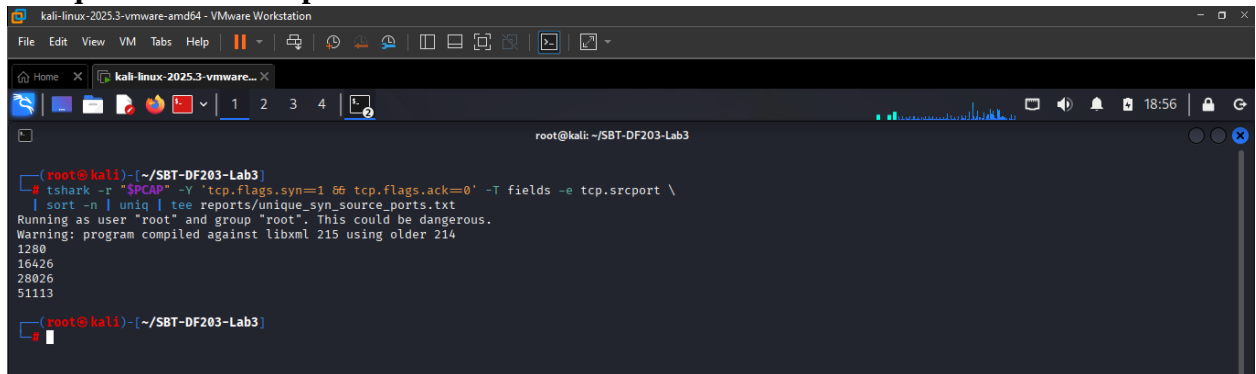
(root@kali)-[~/SBT-DF203-Lab3]
#
```

4 SYN packets

127.0.0.1 -> 127.0.0.1

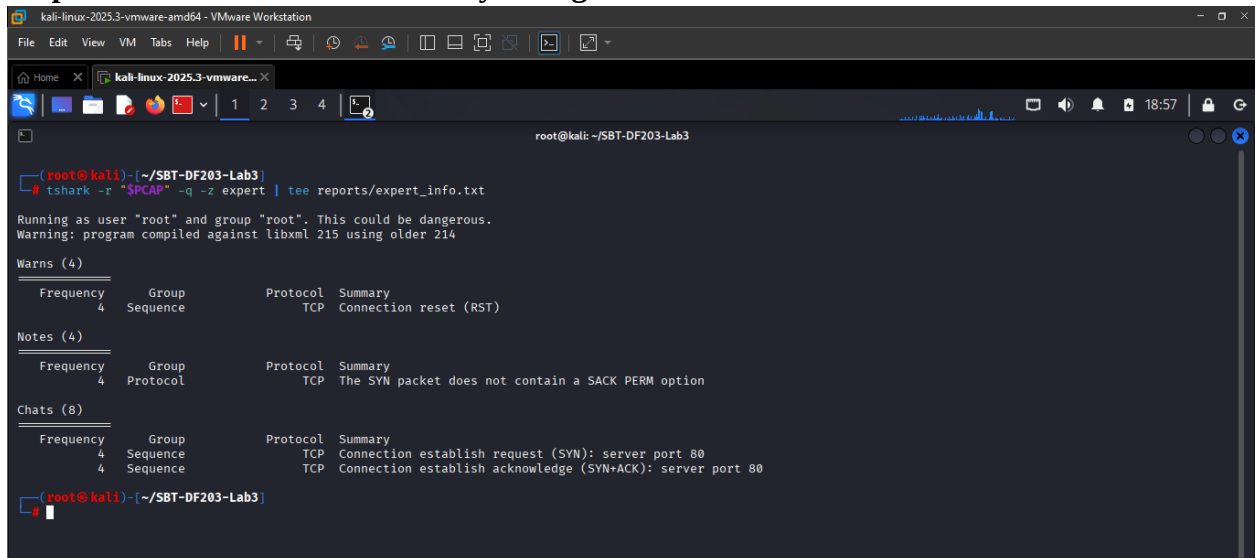
Port 80

2. Unique client source ports



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3
(root@kali)~[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -Y 'tcp.flags.syn==1 && tcp.flags.ack==0' -T fields -e tcp.srcport \
  | sort -n | uniq | tee reports/unique_syn_source_ports.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
1280
16426
28026
51113
(root@kali)~[~/SBT-DF203-Lab3]
```

3. Expert information and TCP analysis flags



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab3
(root@kali)~[~/SBT-DF203-Lab3]
# tshark -r "$PCAP" -q -z expert | tee reports/expert_info.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

Warns (4)
+-----+
| Frequency | Group | Protocol | Summary |
| 4          | Sequence | TCP | Connection reset (RST) |
+-----+

Notes (4)
+-----+
| Frequency | Group | Protocol | Summary |
| 4          | Protocol | TCP | The SYN packet does not contain a SACK PERM option |
+-----+

Chats (8)
+-----+
| Frequency | Group | Protocol | Summary |
| 4          | Sequence | TCP | Connection establish request (SYN): server port 80 |
| 4          | Sequence | TCP | Connection establish acknowledge (SYN+ACK): server port 80 |
+-----+
(root@kali)~[~/SBT-DF203-Lab3]
```

The analysis confirmed repeated SYN activity and allowed correlation of client requests with server responses.

Part E - Compare Normal and Suspicious Sessions

Indicator	Normal HTTP	Bounded SYN Activity
Initial SYN Count	1	4
SYN ACK Count	1	4
Completed Haanshake	Yes	No
Unique Source Ports	1	4
HTTP Request Present	ues	No
RST Packets	Minimal	Present
Duration	Short	short

